

Lab 8 In lab Tasks

Task 1:

Part a)

```
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#include <stdio.h>
#include <stdlib.h>

#define N 10000000

float A[N], B[N], C[N];

int main()
{
    for (int i = 0; i < N; i++)
    {
        A[i] = i * 1.0;
        B[i] = i * 2.0;
    }

    for (int i = 0; i < N; i++)
    {
        C[i] = A[i] + B[i];
    }

    printf("C[0] = %f\n", C[0]);
    printf("C[9999999] = %f\n", C[N-1]);

    return 0;
}
```

```
simroz@simroz: ~/Downloads/lab8
simroz@simroz:~/Downloads/lab8$ touch Q1_partA.c
simroz@simroz:~/Downloads/lab8$ gcc Q1_partA.c -o q1
simroz@simroz:~/Downloads/lab8$ ./q1
C[0] = 0.000000
C[9999999] = 29999996.000000
simroz@simroz:~/Downloads/lab8$
```

Part b)

```
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#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

#define N 10000000
#define THREADS 10

float A[N], B[N], C[N];

void* add_arrays(void* arg)
{
    int id = *(int*)arg;

    int num = N / THREADS;
    int start = id * num;
    int end = (id == THREADS - 1) ? N : start + num;

    for (int i = start; i < end; i++)
    {
        C[i] = A[i] + B[i];
    }

    pthread_exit(NULL);
}
```

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Q1_partB.c
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```
int main()
{
    pthread_t threads[THREADS];
    int ids[THREADS];

    for (int i = 0; i < N; i++)
    {
        A[i] = i * 1.0;
        B[i] = i * 2.0;
    }

    for (int i = 0; i < THREADS; i++)
    {
        ids[i] = i;
        pthread_create(&threads[i], NULL, add_arrays, &ids[i]);
    }

    for (int i = 0; i < THREADS; i++)
    {
        pthread_join(threads[i], NULL);
    }

    printf("C[0] = %f\n", C[0]);
    printf("C[N-1] = %f\n", C[N-1]);

    return 0;
}
```

```
simroz@simroz: ~/Downloads/lab8
simroz@simroz:~/Downloads/lab8$ touch Q1_partB.c
simroz@simroz:~/Downloads/lab8$ gcc Q1_partB.c -o q1B -lpthread
simroz@simroz:~/Downloads/lab8$ ./q1B
C[0] = 0.000000
C[N-1] = 29999996.000000
simroz@simroz:~/Downloads/lab8$
```

Q2:

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Q2.c
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```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>

int *arr;
int n;

float avg;
int min, max;

void* calculate_average(void* arg)
{
    int sum = 0;
    for (int i = 0; i < n; i++)
    {
        sum += arr[i];
    }
    avg = (float)sum / n;
    pthread_exit(NULL);
}

void* calculate_min(void* arg)
{
    min = arr[0];
    for (int i = 1; i < n; i++)
    {
        if (arr[i] < min)
        {
            min = arr[i];
        }
    }
    pthread_exit(NULL);
}
```

```
void* calculate_max(void* arg)
{
    max = arr[0];
    for (int i = 1; i < n; i++)
    {
        if (arr[i] > max)
        {
            max = arr[i];
        }
    }
    pthread_exit(NULL);
}
```

```

int main(int argc, char *argv[])
{
    if (argc < 2)
    {
        printf("Invalid arguments");
        return 1;
    }

    n = argc - 1;

    arr = (int*)malloc(n * sizeof(int));

    for (int i = 0; i < n; i++)
    {
        arr[i] = atoi(argv[i + 1]);
    }

    pthread_t t1, t2, t3;

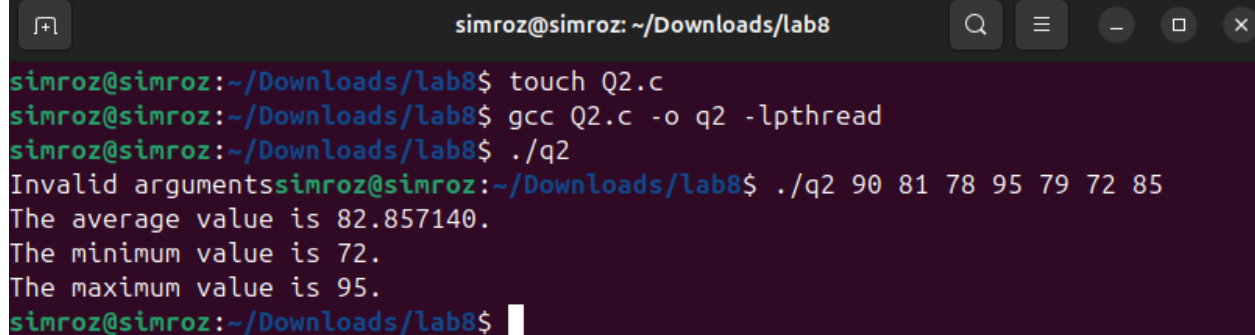
    pthread_create(&t1, NULL, calculate_average, arr);
    pthread_create(&t2, NULL, calculate_min, arr);
    pthread_create(&t3, NULL, calculate_max, arr);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    pthread_join(t3, NULL);

    printf("The average value is %f.\n", avg);
    printf("The minimum value is %d.\n", min);
    printf("The maximum value is %d.\n", max);

    free(arr);
    return 0;
}

```



```

simroz@simroz: ~/Downloads/lab8
simroz@simroz:~/Downloads/lab8$ touch Q2.c
simroz@simroz:~/Downloads/lab8$ gcc Q2.c -o q2 -lpthread
simroz@simroz:~/Downloads/lab8$ ./q2
Invalid arguments
simroz@simroz:~/Downloads/lab8$ ./q2 90 81 78 95 79 72 85
The average value is 82.857140.
The minimum value is 72.
The maximum value is 95.
simroz@simroz:~/Downloads/lab8$

```